QUESTION BANK

GTU EXAM PAPERS (2020 TO 2025)

DISCRETE MATHEMATICS

(OLD – 3140708)

DISCRETE MATHEMATICS AND GRAPH THEORY

(NEW – BE04000261)

MODULE 1 – PROPOSITIONAL AND PREDICATE LOGIC

SR. NO.	TOPIC / QUESTION	YEAR	MARKS
STATEMENTS AND SYMBOLIZATION (PROPOSITIONAL LOGIC)			
1	Symbolize the expression "John is a bachelor and this painting is red".	**JULY 2024**	**03**
2	If p: product is good, q: service is good, r: company is public limited, write the following in symbolic notations (i) either product is good or the service is good or both, (ii) either the product is good or service is good but not both, (iii) it is not the case that both product is good and company is public limited.	**OCT 2020**	**03**
EQUIVALENCE OF FORMULAS (PROPOSITIONAL LOGIC)			
1	Show that the propositions $p \rightarrow q$ and $\sim p \vee q$ are logically equivalent.	**NOV 2025**	**03**
2	Prove that $p \rightarrow q$ and $(\neg p \vee q)$ are logically equivalent.	**JULY 2023**	**03**
3	Show that the propositions $\neg (p \wedge q)$ and $\neg p \vee q$ are logically equivalent.	**FEB 2021**	**07**
TAUTOLOGY			
1	Show that the proposition $\big(p \rightarrow (q \rightarrow r)\big) \rightarrow \big((p \rightarrow q) \rightarrow (p \rightarrow r)\big)$ is a tautology.	**NOV 2025**	**04**
2	Show that the proposition $\lfloor (p \rightarrow q) \wedge (q \rightarrow r) \rfloor \rightarrow (p \rightarrow r)$ is a tautology.	**MAY 2025**	**04**
TAUTOLOGIES AND EQUIVALENCE OF FORMULAS (PROPOSITIONAL LOGIC)			
1	Identify the statement $(p \rightarrow q) \leftrightarrows (\neg p \vee q)$ is tautology or contradiction.	**NOV 2024**	**04**
2	Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.	**FEB 2021**	**03**
TAUTOLOGIES AND LOGICAL EQUIVALENCE (PROPOSITIONAL LOGIC)			
1	Identify the statement $\big(\neg q \wedge (p \rightarrow q)\big) \rightarrow \neg p$ is tautology or contradiction without constructing the truth table.	**JULY 2024**	**03**
TAUTOLOGIES, EQUIVALENCE OF FORMULAS, AND TRUTH TABLES (PROPOSITIONAL LOGIC)			
1	Check whether the proposition $\big((p \vee q) \wedge \sim p\big) \rightarrow q$ is contradiction, tautology or contingency.	**NOV 2025**	**03**

2	Identify the statement $(p \wedge q) \wedge \neg (p \vee q)$ is tautology or contradiction.	**FEB 2024**	**03**
3	Define Functionally complete set of connectives, Principal Disjunctive Normal Form (PDNF). Obtain PDNF for the expression $\neg \big((p \wedge q) \vee (\neg p \wedge q) \vee (q \wedge r)\big)$	**DEC 2022**	**07**
4	(1) Use a truth table to verify the De Morgan's law $\neg (p \vee q) \equiv \neg p \wedge \neg q$ (2) Show that $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology.	**JULY 2022**	**07**
5	Find if the following is a tautology, contradiction or contingency. $\big(p \rightarrow (q \rightarrow r)\big) \rightarrow \big((p \rightarrow q) \rightarrow (p \rightarrow r)\big)$	**SEP 2021**	**07**
6	Obtain the d.n.f. of the form $(p \rightarrow q) \wedge (\neg p \wedge q)$.	**FEB 2021**	**04**
7	Obtain the d.n.f. of the form $\neg \big(p \rightarrow (q \wedge r)\big)$.	**OCT 2020**	**04**
TAUTOLOGICAL IMPLICATIONS / VALIDITY OF ARGUMENTS (PROPOSITIONAL LOGIC)			
1	Use a truth table to determine whether the following argument form is valid. $p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$	**NOV 2024** **FEB 2024**	**07** **07**
2	Use a truth table to determine whether the following argument form is valid. $p \rightarrow q$ $p \rightarrow r$ $\therefore p \rightarrow q \vee r$	**JULY 2024**	**07**
3	Using truth table state whether each of the following implication is tautology. (a) $(p \wedge r) \rightarrow p$ (b) $(p \wedge q) \rightarrow (p \rightarrow q)$ (c) $p \rightarrow (p \vee q)$	**DEC 2021**	**07**
4	Determine the validity of the argument given: If I study, then I will not fail in Discrete Mathematics. If I do not play cricket, then I will study. But I failed in Discrete Mathematics. --- Therefore, I must have played cricket.	**SEP 2021**	**07**
5	Find out using truth table, whether $(p \wedge r) \rightarrow p$ is a tautology.	**OCT 2020**	**03**
QUANTIFIERS AND NEGATION OF STATEMENTS (PREDICATE LOGIC)			
1	Negate (opposite) each of the following statements:	**NOV 2025**	**04**

	(i) $\forall\, x\ ,\ |x| = x$ (ii) $\exists\ x\ ,\ x^2 = x$ (iii) If there is a riot, then someone killed. (iv) It is day light and all the people are arisen.		
PREDICATE FORMULAS AND SYMBOLIZATION (PREDICATE LOGIC)			
1	Symbolize the following expressions "x is the father of the mother of y" where $P(x) : x$ is a person, $F(x, y) : x$ is the father of y and $M(x, y) : x$ is the mother of y.	**NOV 2024**	**03**
QUANTIFIERS AND LOGICAL INFERENCE / VALID ARGUMENTS (PREDICATE LOGIC)			
1	Show that the premises "Everyone in this discrete mathematics class has taken a course in computer science" and "Marla is a student in this class" imply the conclusion "Marla has taken a course in computer science."	**NOV 2024**	**04**
QUANTIFIERS, LOGICAL CONNECTIVES, AND VALIDITY OF ARGUMENTS (PREDICATE LOGIC)			
1	Express the following using predicate, quantifier and logical connectives. Also verify the validity of the consequence. Everyone who graduates gets a job. Ram is graduated. Therefore, Ram got a job.	**JULY 2024**	**04**
2	Express the following using predicate, quantifier and logical connectives. Also verify the validity of the consequence. (i) Every computer science major takes discrete mathematics. Natasha is taking discrete mathematics. Therefore, Natasha is a computer science major. (ii) All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.	**FEB 2024**	**07**
QUANTIFIERS AND TRUTH VALUES OF PREDICATES (PREDICATE LOGIC)			
1	Let $P(x)$ be the statement " $x = x^2$ ". If the domain consists of the integers, what are the truth values of the following? (1) $\exists x\, P(x)$ (2) $\forall x \neg P(x)$ (3) $\exists x \neg P(x)$	**JULY 2022**	**03**
QUANTIFIERS, RULES OF INFERENCE, AND VALIDITY OF ARGUMENTS (PREDICATE LOGIC)			
1	Identify the error or errors in this argument that supposedly shows that if $\exists x\, P(x) \wedge \exists x\, Q(x)$ is true then $\exists\, x\, (P(x) \wedge Q(x))$ is true. (1) $\exists x\, P(x) \wedge \exists x\, Q(x)$ Premise (2) $\exists x\, P(x)$ Simplification from (1) (3) $P(c)$ Existential instantiation from (2) (4) $\exists x\, Q(x)$ Simplification from (1) (5) $Q(c)$ Existential instantiation from (2)	**JULY 2022**	**04**

	(6) $P(c) \wedge Q(c)$ Conjunction from (3) and (5) (7) $\exists\, x\,(P(x) \wedge Q(x))$ Existential generalization		
QUANTIFIERS, DOMAINS, AND LOGICAL EQUIVALENCE OF QUANTIFIED STATEMENTS (PREDICATE LOGIC)			
1	(1) Suppose that the domain of $Q(x, y, z)$ consists of triples x, y, z where $x = 0, 1$ or 2 , $y = 0$ or 1 , and $z = 0$ or 1. Write out following propositions using disjunctions and conjunctions. (i) $\exists\, z \neg Q(0, 0, z)$ (ii) $\forall\, y\, Q(0, y, 0)$ (2) (i) Show that $\exists\, x\, P(x) \wedge \exists\, x\, Q(x)$ and $\exists\, x\,(P(x) \wedge Q(x))$ are not logically equivalent. (ii) Show that $\exists\, x\,(P(x) \vee Q(x))$ and $\exists\, x\, P(x) \vee \exists\, x\, Q(x)$ are logically equivalent.	**JULY 2022**	**07**
QUANTIFIERS: EXISTENTIAL QUANTIFIER (PREDICATE LOGIC)			
1	Define: Existential Quantifier	**SEP 2021**	**01**
QUANTIFIERS AND SYMBOLIZATION OF STATEMENTS (PREDICATE LOGIC)			
1	Rewrite the following statements using quantifier variables and predicate symbols: 1) All birds can fly. 2) Some women are genius. 3) There is a student who likes Discrete Mathematics but not Probability and Statistics. 4) Each integer is either even or odd.	**SEP 2021**	**04**
QUANTIFIERS AND LOGICAL SYMBOLIZATION (PREDICATE LOGIC)			
1	For the universe of integers, let $p(x)$, $q(x)$, $r(x)$, $s(x)$ and $t(x)$ be the following open statements: $p(x)$: $x > 0$; $q(x)$: x is even ; $r(x)$: x is a perfect square ; $s(x)$: x is divisible by 4 ; $t(x)$: x is divisible by 5. Write the following statements in symbolic form: (i) At least one integer is even, (ii) There exists a positive integer that is even, (iii) If x is even then x is not divisible by 5, (iv) No even integer is divisible by 5, (v) There exists an even integer divisible by 5.	**OCT 2020**	**04**
FREE & BOUND VARIABLES, QUANTIFIERS, AND LOGICAL INFERENCE / VALIDITY OF ARGUMENTS (PREDICATE LOGIC)			
1	Define free variable and bound variable. Rewrite the following argument using quantifiers, variables and predicate symbols. Prove the validity of the argument. "All healthy people eat an apple a day. Ram does not eat apple a day. Ram is not a healthy person."	**DEC 2022**	**07**

MODULE 2 – SET THEORY

SR. NO.	TOPIC / QUESTION	YEAR	MARKS
INCLUSION (SUBSET)			
1	Determine whether each of these statements is true or false. (1) $0 \in \emptyset$ (2) $\emptyset \subset \{0\}$ (3) $\{0\} \in \{0\}$ (4) $\emptyset \in \{\emptyset\}$ (5) $\{\emptyset\} \in \{\{\emptyset\}\}$ (6) $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$	**JULY 2022**	**03**
POWER SET			
1	Find the power sets of (i) $\{a\}$, (ii) $\{a, b, c\}$.	**FEB 2021**	**03**
UNION AND INTERSECTION AND DIFFERENCE			
1	Given $A = \{x : x$ is an integer and $1 \leq x \leq 5\}$, $B = \{3, 4, 5, 17\}$ and $C = \{1, 2, 3, \ldots\}$, find $A \cap B$, $A \cup B$, $A \cap B \cap C$ and $A \cup C$.	**FEB 2024**	**04**
2	If $A = \{4, 5, 7, 8, 10\}$, $B = \{4, 5, 9\}$ and $C = \{1, 4, 6, 9\}$, then verify that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.	**DEC 2022**	**04**
3	If $S = \{a, b, c\}$, find nonempty disjoint sets A_1 and A_2 such that $A_1 \cup A_2 = S$. Find the other solutions to this problem.	**DEC 2021**	**04**
4	If $A = \{x \mid x = 3^n,\ n \leq 6,\ n \in N\}$ and $B = \{x \mid x = 9^n,\ n \leq 4,\ n \in N\}$ then find $A \cup B$, $A \cap B$ and $A - B$.	**JULY 2023**	**03**
5	Given $A = \{2, 5, 6\}$, $B = \{3, 4, 2\}$, find $(A - B)$ and $(B - A)$.	**JULY 2024**	**02**
CARTESIAN PRODUCT			
1	If $A = \{a, b\}$ and $B = \{c, d\}$ and $C = \{e, f\}$ then find (i) $(A \times B) \cup (B \times C)$ (ii) $A \times (B \cup C)$.	**OCT 2020**	**03**
INCLUSION–EXCLUSION PRINCIPLE			
1	In a class of 50 students, 12 enrolled for both Mathematics and Science, 32 enrolled for science. If the students of the class enrolled for at least one of the two subjects, then how many students enrolled for only Mathematics but not Science?	**FEB 2024**	**04**
2	In a survey of 400 students in a school, 100 were found as drinking apple juice, 150 as drinking orange juice and 75 drinking both apple and orange juice. Find how many students drink neither apple nor orange juice.	**JULY 2023**	**03**
3	Suppose that 100 mathematics students at a college taking at least one of the languages French, German, and Russian, given the following data:	**NOV 2024**	**04**

	65 study French, 45 study German, 42 study Russian, 20 study French and German, 25 study French and Russian, 15 study German and Russian. (1) Find the number of students who study all the three languages. (2) Find the number of students who study only French.		
4	How many integers between 1 to 2000 are divisible by $2, 3, 5$ or 7?	**NOV 2025**	**07**
VENN DIAGRAMS			
1	Let A, B and C be the sets such that $(A \cap B \cap C) = \emptyset$, $A \cap B = \emptyset$, $A \cap C = \emptyset$, $B \cap C = \emptyset$. Draw the corresponding Venn diagram.	**NOV 2025**	**03**
SET THEORY IDENTITIES			
1	Using Venn diagram, prove the following: (i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (ii) $A \oplus (B \oplus C) = (A \oplus B) \oplus C$	**NOV 2025**	**07**
2	Prove that: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	**JULY 2024**	**04**
3	Prove that $(A \cap B) \cup (A - B) = A$ and $A \cup (B - A) = A \cup B$	**MAY 2025**	**03**
4	Show that for any two sets A and B , $A - (A \cap B) = A - B$.	**DEC 2021**	**03**
5	Prove that $(A \cup B)' \equiv A' \cap B'$.	**SEP 2021**	**04**
6	Prove that $A \cap \bar{B} = A \cap \bar{C}$ if and only if $A \cap B = A \cap C$.	**NOV 2024**	**04**

MODULE 2 – COUNTING AND COMBINATORICS

SR. NO.	TOPIC / QUESTION	YEAR	MARKS
PIGEONHOLE PRINCIPLE			
1	Among 100 people at least how many of them were born in the same month?	**SEP 2021**	**03**
PERMUTATIONS			
1	(i) Suppose repetitions are not allowed, how many four-digit numbers can be formed from six digits $1, 2, 3, 5, 7, 8$? (ii) How many of such numbers less than 4000? (iii) How many in (i) are even? (iv) How many in (i) are divisible by 10?	**JULY 2024**	**04**
COMBINATIONS			
1	A history class contains 8 male students and 6 female students. Find the number n of ways that the class can elect: (a) 1 class representative (b) 2 class representatives, 1 male and 1 female (c) 1 president and 1 vice president.	**NOV 2024**	**03**
2	In a box, there are 5 black pens, 3 white pens and 4 red pens. In how many ways can 2 black pens, 2 white pens and 2 red pens can be chosen?	**DEC 2021** **FEB 2021**	**04**
3	A committee of 5 persons, is to be formed from 6 men and 4 women. In how many ways this can be done when (i) at least 2 women are included, (ii) at most 2 women are included.	**DEC 2022** **OCT 2020**	**03** **04**
4	From 4 professors and 6 students, a committee of 3 is to be formed. In how many ways, this can be done, if the committee contains **(1)** at most 1 professor **(2)** at least 2 professors.	**JULY 2023**	**04**
5	The Indian cricket team consist of 16 players. It includes 2 wicket keepers and 5 bowlers. In how many ways eleven player can be selected if we have to select one wicket keeper and at least 4 bowlers?	**JULY 2023** **DEC 2021**	**03** **04**
6	Find total number of diagonals in a polygon with n sides.	**MAY 2025**	**03**
BINOMIAL COEFFICIENTS AND BINOMIAL THEOREM			
1	Find n if $nC_3 + (n+2)C_3 = nP_3$.	**JULY 2023**	**03**
MATHEMATICAL INDUCTION			
1	Prove that $(1 \times 2) + (2 \times 3) + \cdots + \big(n \times (n+1)\big) = \dfrac{n(n+1)(n+2)}{3}$	**MAY 2025**	**03**

MODULE 2 – RECURRENCE RELATION

SR. NO.	TOPIC / QUESTION	YEAR	MARKS
HOMOGENEOUS RECURRENCE RELATION			
1	Solve the following recurrence relations: (i) $a_n - 7a_{n-1} + 10a_{n-2} = 0$ given that $a_0 = 0$, $a_1 = 3$ (ii) $a_n - 4a_{n-1} + 4a_{n-2} = 0$ given that $a_0 = 1$, $a_1 = 6$	**NOV 2025**	**07**
2	Solve: $a_n = 2a_{n-1} + 3a_{n-2}$, $a_0 = 1$, $a_1 = 2$	**NOV 2024**	**04**
3	Solve $a_n = a_{n-1} + a_{n-2}$, $a_0 = 0$, $a_1 = 1$	**JULY 2024**	**04**
4	Solve the recurrence relation which represents the Fibonacci sequence $F_n = F_{n-1} + F_{n-2}$ with $F_0 = F_1 = 1$.	**JULY 2023**	**07**
5	Solve: $a_n = 11a_{n-1} - 39a_{n-2} + 45a_{n-3}$, where $a_0 = 5$, $a_1 = 11$, $a_2 = 25$.	**FEB 2024**	**07**
NONHOMOGENEOUS RECURRENCE RELATION			
1	Solve the recurrence relation $a_n - 3a_{n-1} = 2, n \geq 1, a_0 = 1$.	**NOV 2025**	**04**
2	Solve the recurrence relation $S_n - 6S_{n-1} + 9S_{n-2} = 3^{n+1}$	**MAY 2025**	**07**
3	Solve the recurrence relation $a_n + 5a_{n-1} + 6a_{n-2} = 3n^2$ using the method of undetermined coefficients.	**OCT 2020**	**07**
4	Solve the recurrence relation $a_{n+2} - 5a_{n+1} + 6a_n = 2$.	**SEP 2021**	**07**
5	Solve the recurrence relation $a_{n+2} - 5a_{n+1} + 6a_n = 2$ with initial condition $a_0 = 1$ and $a_1 = -1$ using method of undetermined coefficients.	**DEC 2022**	**07**
6	Solve the recurrence relation $a_n - 4a_{n-1} + 4a_{n-2} = n + 3^n$ using method of undetermined coefficients.	**FEB 2021**	**07**
7	Solve the recurrence relation $S_n + 3nS_{n-1} = 0, S_0 = 1$	**MAY 2025**	**07**
METHOD OF GENERATING FUNCTION			
1	Solve the recurrence relation using the method of generating function $a_n - 5a_{n-1} + 6a_{n-2} = 3n$, $n \geq 2$, $a_0 = 0$, $a_1 = 2$.	**NOV 2024**	**07**
2	Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 3^n$, $n \geq 2$, $a_0 = 0$, $a_1 = 2$ using the method of generating function.	**OCT 2020**	**07**
3	Use generating function to solve a recurrence relation $a_n = 3a_{n-1} + 2$ with $a_0 = 1$.	**DEC 2022**	**07**
4	Find the generating function of $a_{n+2} = a_{n+1} + a_n$ where $a_0 = a_1 = 1$ for $n \geq 1$.	**JULY 2023**	**07**

MODULE 3 – FUNCTIONS

SR. NO.	TOPIC / QUESTION	YEAR	MARKS
INJECTIVE (ONE-ONE) & SURJECTIVE (ONTO) FUNCTION			
1	Define onto function. Check whether the function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is one-one and onto.	**NOV 2024**	**03**
2	Define injective function.	**JULY 2024**	**01**
3	A function $f: \mathbb{R}^+ \to \mathbb{R}$ is defined by $f(x) = x^2 - 8$. Check whether f is one-one and onto.	**FEB 2024**	**03**
BIJECTIVE FUNCTION			
1	Show that the function $f: R \to R$ such that $f(x) = x^3 + x$ is a bijection.	**MAY 2025**	**04**
2	Define even and odd functions. Determine whether the function $f: R \to R^+$ defined by $f(x) = 2x + 7$ is one-to-one or bijective.	**OCT 2020**	**04**
COMPOSITION OF FUNCTIONS			
1	Define composition of functions.	**SEP 2021**	**01**
2	If $f(x) = \frac{x+1}{x-1}$ and $g(x) = \frac{x-1}{x+1}$ then find $gof(x)$ and $fog(x)$.	**MAY 2025**	**04**
3	Find the composition of two functions $f(x) = e^x$ and $g(x) = x^3$, $(f \ o \ g)(x)$ and $(g \ o \ f)(x)$. Hence, show that $(f \ o \ g)(x) \neq (g \ o \ f)(x)$.	**DEC 2021**	**03**
4	If $f(x) = 2x$, $g(x) = x^2$, $h(x) = x + 1$ then find $(f \ o \ g) \ o \ h$ and $f \ o \ (g \ o \ h)$.	**FEB 2021**	**04**
5	If $f: x \to 2x$, $g: x \to x^2$ and $h: x \to (x + 1)$, then show that $(f \ o \ g) o \ h = f \ o (g \ o \ h)$.	**OCT 2020**	**03**
6	Let $f: R \to R$ be function defined by $f(x) = ax + b$. Find the values of a and b for which $f \ o \ f = I_R$.	**JULY 2023**	**04**
7	Show that the functions $f(x) = x^3$ and $g(x) = x^{1/3}$ for $x \in \mathbb{R}$ are inverses of one another.	**FEB 2024**	**03**
INVERSE OF FUNCTIONS			
1	Check whether the function $f(x) = x^3 - 2$, for $x \in \mathbb{R}$ is invertible function. If so, find $f^{-1}(x)$.	**JULY 2024**	**03**

2	Define one - one function. Show that the function $f : R \to R$, $f(x) = 3x - 7$ is one - one and onto both. Also, find its inverse.	**DEC 2022**	**04**
3	Let $Y = \{n^2 \ : \ n \in N\}$ and $f: N \to Y$ defined as $f(n) = n^2$. Show that f is invertible and find the inverse of f.	**JULY 2023**	**04**
4	Show that the function $f: R \to R^+ \cup \{0\}$ defined by $f(x) = \|x\|$ is not invertible. Modify the domain or codomain of f so that it becomes invertible.	**JULY 2022**	**03**
BASIC CONCEPT OF FUNCTION & RANGE			
1	Determine whether f is a function from the set of all bit strings to the set of integers if 1) $f(s)$ is the position of a 0 bit in s. 2) $f(s)$ is the number of a 1 bits in s. Find the range of each of the following functions that assigns: 3) to a bit string the number of one bits in the string 4) to each bit string twice the number of zeros in that string.	**JULY 2022**	**04**
CHARACTERISTIC FUNCTION AND ALGEBRA			
1	Let S be subset of a universal set U. The characteristic function $f_S : U \to \{0,1\}$, $f_S(x) = 1$ if $x \in S$ and 0 if $x \notin S$. Let A and B be sets. Then, show that $f_{A \cap B}(x) = f_A(x) \cdot f_B(x)$.	**JULY 2022**	**03**
EVEN AND ODD FUNCTION			
1	Let $f(x)$ be any real valued function. Show that $g(x) = \frac{f(x)+f(-x)}{2}$ is always an even function where as $h(x) = \frac{f(x)-f(-x)}{2}$ is always an odd function.	**DEC 2021**	**03**

MODULE 3 – ALGEBRAIC STRUCTURES

SR. NO.	TOPIC / QUESTION	YEAR	MARKS
BINARY OPERATION			
1	Is addition a binary operation on $\{-1, 0, 1\}$? Justify.	**SEP 2021**	**04**
COMPOSITE TABLE			
1	Draw the composite table for the operation $*$ defined by $x * y = x, \forall\, x, y \in S = \{a, b, c, d\}$.	**FEB 2021**	**03**
2	Construct the composite tables for **(i)** addition modulo 4 and **(ii)** multiplication modulo 4 for $Z_4 = \{0, 1, 2, 3\}$. Check whether they have identity and inverse element.	**OCT 2020**	**03**
PROPERTIES OF COMMUTATIVE AND ASSOCIATIVE			
1	Check the properties of commutative and associative for the operation $*$ defined by $x * y = x + y - 2$ on the set Z of integers.	**FEB 2021**	**03**
2	Show that the operation $*$ defined by $x * y = x^y$ on the set N of natural numbers is neither commutative nor associative.	**OCT 2020** **FEB 2021**	**03** **03**
IDENTITY ELEMENT			
1	Consider the set Q of rational numbers. Let $*$ be the operation on Q defined by $a * b = a + b - ab$. Find **(i)** $3 * 4$ **(ii)** the identity element for $*$.	**JULY 2022**	**03**
ALGEBRAIC STRUCTURE, SEMIGROUP AND MONOID			
1	(i) Define: Semigroups (ii) Let $S = \{a, b, c, d\}$. The following multiplication table, define operation $\cdot$ on S. Is $\langle S\ ,\ \cdot \rangle$ semigroup? Justify. $\begin{array}{c\|cccc} \cdot & a & b & c & d \\ \hline a & a & b & c & d \\ b & b & a & a & b \\ c & c & b & a & a \\ d & d & a & a & a \end{array}$	**JULY 2022**	**03**
2	Define: Monoid	**SEP 2021**	**01**
3	Define algebraic structure, semi group and monoid. Also give related examples.	**OCT 2020**	**03**

4	Define Semi Group, Monoid. Give an example of an algebraic structure which is semi group but not monoid.	**DEC 2022**	**03**
5	Consider the set of positive integers $\mathbb{N}$. Check which of $(\mathbb{N}, +)$ and $(\mathbb{N}, \times)$ are semigroup and which are monoid?	**JULY 2024**	**04**
GROUP **GROUP (OPERATION MULTIPLICATION)**			
1	Is the algebraic system $(Q\ ,\ *)$ a group? Where Q is the set of rational numbers and $*$ is a binary operation defined by $a * b = a + b - ab, \forall a, b \in Q$.	**OCT 2020**	**03**
2	Let $G = \{(a, b) \mid a, b \in R\}$. Define binary operation $(*)$ on G as $(a, b), (c, d) \in G$, $(a, b) * (c, d) = (ac\ , bc + d)$. Show that an algebraic structure $(G\ ,\ *)$ is a group.	**DEC 2021**	**07**
3	Show that the algebraic structure $(G\ ,\ *)$ is a group, where $G = \{(a, b) \mid a, b \in R, a \neq 0\}$ and $*$ is a binary operation define by $(a, b) * (c, d) = (ac\ , bc + d)$ for all $(a, b), (c, d) \in G$.	**FEB 2021**	**03**
4	Show that the set $\{1\ , -1\ , i\ , -i\}$ is a group with respect to multiplication.	**MAY 2025**	**03**
GROUP - THEOREMS			
1	If $(G,*)$ be a group then for any two elements a and b of $(G,*)$, prove that $(a * b)^{-1} = b^{-1} * a^{-1}$.	**FEB 2024**	**02**
2	Prove that the inverse of an element is unique in a group $(G,\ *)$.	**FEB 2021**	**04**
ABELIAN GROUP **ABELIAN GROUP (OPERATION ADDITION MODULO)** **ABELIAN GROUP (OPERATION MULTIPLICATION MODULO)** **ABELIAN GROUP (OPERATION MATRIX ADDITION AND MULTIPLICATION)**			
1	Define Abelian group and prove that the set $\{0,1,2,3,4\}$ is a finite abelian group under addition modulo 5 as binary operation.	**DEC 2022**	**03**
2	Prove that $G = \{1, 2, 3, 4, 5, 6\}$ is a finite abelian group of order 6 with respect to multiplication modulo 7.	**SEP 2021**	**07**
3	Let G be the set of non-zero real numbers. Define binary operation $(*)$ on G as $a * b = \frac{ab}{2}$. Show that an algebraic structure $(G\ ,\ *)$ is an abelian group.	**DEC 2021**	**07**
4	Prove that an algebraic structure $(G\ ,\ *)$ is an abelian group, where G is the set of non-zero real numbers and $*$ is a binary operation defined by $a * b = \frac{ab}{2}$.	**OCT 2020**	**04**

5	Show that G is an abelian group under usual matrix addition where $G = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \middle\| \; a, b, c, d \in R \right\}$.	**JULY 2023**	**07**
6	Show that an algebraic structure $(G, \cdot)$ is an abelian group, where $G = \{A, B, C, D\}$, $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, D = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ and $\cdot$ is the binary operation of matrix multiplication.	**FEB 2021**	**04**
7	Define abelian group for any non-empty set.	**NOV 2025**	**03**
8	Let Q_+ be the set of all positive rational numbers and $*$ be a binary operation on Q_+ defined by $a * b = ab/3$ then show that $(Q_+, *)$ is an abelian group.	**NOV 2025**	**03**
9	If $G = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and operation $+_8$ is an addition modulo 8, then show that $(G, +_8)$ is an abelian group.	**NOV 2025**	**04**
10	Let $\mathbb{Z}_m$ denotes the integers modulo m. Check whether $\mathbb{Z}_m$ is a group under addition. If so, is it abelian group?	**NOV 2024**	**04**
11	Let G be the subset of 2×2 real matrices with a nonzero determinant. Check whether G is group under matrix multiplication. If so, is it abelian group?	**JULY 2024**	**04**
ABELIAN GROUP - THEOREMS			
1	If G is an abelian group with n elements $g_1, g_2, \ldots, g_n$ then show that $(g_1 g_2 \ldots g_n)^2 = e$ where e is the identity element of G.	**MAY 2025** **JULY 2023**	**04** **03**
2	Show that if every element in a group is its own inverse, then the group must be abelian.	**NOV 2024**	**03**
3	Show that in a group $(G,*)$, for any $a, b \in G$ if $(a * b)^2 = a^2 * b^2$ then $(G,*)$ must be abelian.	**FEB 2024**	**02**
NON-ABELIAN GROUP			
1	Define group with example. Give an example of a non-abelian group.	**SEP 2021**	**03**
SUBGROUP **SUBGROUP (OPERATION MATRIX MULTIPLICATION)**			
1	Define subgroup of a group. Also, show that the subset H of a group of integers I is a subgroup under addition, where $H = \{\ldots, -2m, -m, 0, m, 2m, \ldots\}$.	**NOV 2025**	**04**

2	Show that arbitrary intersection of subgroups of a group is a subgroup of a group. Will union of two subgroups be also subgroup of group?	**MAY 2025**	**07**
3	Show that $H = \left\{\begin{bmatrix} 1 & 0 \\ a & 1 \end{bmatrix} : a \in R\right\}$ is a subgroup of G where G is the group of all non-singular lower triangular matrices of order 2×2 over R with usual matrix multiplication.	**MAY 2025**	**03**
4	Define subgroup and group Homomorphism.	**SEP 2021**	**03**
NORMAL SUBGROUP			
1	The subset $H = \{0, 2\}$ is a subgroup of $(Z_4, +_4)$. Is H a normal subgroup?	**NOV 2024**	**03**
2	Show that every subgroup of a cyclic group is normal.	**FEB 2024**	**03**
3	The subset $H = \{0,2\}$ is a subgroup of $(\mathbb{Z}_4\ ,\ +_4)$. Find all left and right cosets of H in $(\mathbb{Z}_4\ ,\ +_4)$. Is H a normal subgroup?	**FEB 2024**	**04**
4	Prove that every subgroup H of an abelian group is normal.	**OCT 2020**	**04**
LEFT COSETS AND RIGHT COSETS			
1	Find left cosets and right cosets of $H = \{0,3\}$ in the group $(\mathbb{Z}_6, +_6)$.	**JULY 2024**	**03**
2	Show that the only right coset of a subgroup H in a group G that is also subgroup of G is H itself.	**JULY 2023**	**03**
3	Let $H = \{[0], [3]\}$ in Z_6 under addition. Find left and right cosets in $\langle Z_6\ , +_6 \rangle$.	**SEP 2021**	**04**
4	Write the equivalence classes for congruence modulo 4 i.e. z_4. Let the subset $H = \{[0], [2]\}$ is a subgroup of $G = \langle z_4\ , +_4 \rangle$. Then, determine all left cosets of H in G.	**JULY 2022**	**04**
5	Let $H = \{1, -1\}$ and $G = \{1, -1, i, -i\}$. $(H,\times)$ is a sub-group of $(G,\times)$. Then, find all left cosets and right cosets of H in G.	**JULY 2022**	**04**
6	Define subgroup. Let H be a subgroup of $(Z, +)$ where H is the set of even integers and Z is the set of all integers and $+$ is the operation of addition. Find all right cosets of H in Z.	**FEB 2021**	**03**
7	Let $(Z, +)$ be a group, where Z is the set of integers and $+$ is an operation of addition. Let H be a subgroup of Z consisting of elements multiple of 5. Find the left cosets of H in Z.	**OCT 2020**	**04**
CYCLIC GROUP			
1	Define cyclic group and show that the following groups are cyclic groups:	**NOV 2025**	**04**

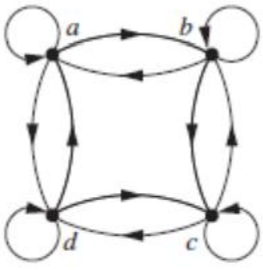
	(i) Third roots of unity i.e. $G = \{1, \omega, \omega^2\}$, $\omega^3 = 1$ (ii) $G = \{1, -1, i, -i\}$ where $i^2 = -1$		
2	Find all the generators of cyclic group $(Z_5, +_5)$.	**NOV 2024**	**03**
3	Prove that every cyclic group is abelian.	**JULY 2024**	**03**
4	Show that $(\mathbb{Z}_5^*, \times_6)$ is cyclic group where $\mathbb{Z}_5^* = \mathbb{Z}_5 \backslash \{0\}$.	**FEB 2024**	**03**
5	Show that $(G, +_5)$ is a cyclic group, where $G = \{0, 1, 2, 3, 4\}$.	**OCT 2020**	**03**
6	Define Cyclic group, Normal subgroup. Illustrate with an example.	**DEC 2022**	**03**
LAGRANGE'S THEOREM			
1	State Lagrange's theorem and find out all possible subgroups of group $[\{1, -1, i, -i\}, \times]$.	**DEC 2022**	**03**
2	Explain Cosets and Lagrange's theorem.	**SEP 2021**	**07**
HOMOMORPHISM			
1	Define homomorphism. Let G be the group of real numbers under addition, and let G' be the group of positive real numbers under multiplication. Check whether the mapping $f: G \to G'$ defined by $f(a) = 2^a$ is a homomorphism.	**NOV 2024**	**03**
2	Let g be a homomorphism from a group $(G, *)$ to a group (H, Δ). Show that $g(e_G) = e_H$ and for any $a \in G$, $g(a^{-1}) = \big(g(a)\big)^{-1}$.	**JULY 2024**	**03**
3	Define $f: (\mathbb{N} \times \mathbb{N}, *) \to (\mathbb{Q}, \times)$ by $f(a, b) = \frac{a}{b}$. Show that f is a homomorphism.	**FEB 2024**	**03**
GROUP PERMUTATION			
1	Define group permutation. Find the inverse of the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix}$.	**FEB 2021**	**04**
2	Show that the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 6 & 2 & 4 & 1 & 3 \end{pmatrix}$ is odd and $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 3 & 4 & 5 & 2 & 1 \end{pmatrix}$ is even.	**DEC 2022**	**04**
RING (ALGEBRAIC STRUCTURES WITH TWO BINARY OPERATIONS)			
1	Consider the ring $\mathbb{Z}_{30} = \{0, 1, 2, \ldots, 29\}$ of integers modulo 30. (a) Find -8 and -17. (b) Find 11^{-1}, 13^{-1} and 14^{-1}. (c) Let $f(x) = 2x + 4$. Find the roots of $f(x)$ over $\mathbb{Z}_{30}$.	**NOV 2024**	**07**

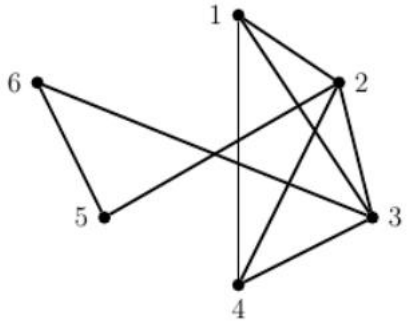
2	Define ring. Show that the algebraic system $(Z_9, +_9, \cdot_9)$, where $Z_9 = \{0, 1, 2, 3, \ldots, 8\}$ under the operations of addition and multiplication of congruence modulo 9, form a ring.	**FEB 2021**	**04**
3	Define ring. Show that the set $M = \left\{ \begin{bmatrix} a & 0 \\ b & 0 \end{bmatrix} \mid a, b \in R \right\}$ is not a ring under the operations of matrix addition and multiplication.	**OCT 2020**	**04**
4	Define Ring. Write elements of the ring $\langle z_{10}, +_{10}, \times_{10} \rangle$. Find $-3, -8, 3^{-1}, 4^{-1}$.	**JULY 2022**	**07**
5	Consider the ring $\mathbb{Z}_{10} = \{0, 1, 2, 3, \ldots, 9\}$ of integers modulo 10. (a) Find the units of $\mathbb{Z}_{10}$. (b) Find $-3, -8$ and 3^{-1}. (c) Let $f(x) = 2x^2 + 4x + 4$. Find the roots of $f(x)$ over $\mathbb{Z}_{10}$.	**FEB 2024**	**07**
6	We are given the ring $\langle \{a, b, c, d\}, +, \cdot \rangle$ the operations $+$ and $\cdot$ on R are as shown in the following table: $+$: a a b c d / b b c d a / c c d a b / d d a b c $\cdot$: a a a a a / b a c a c / c a a a a / d a c a a (i) Does it have an identity? (ii) What is the zero of this ring? (iii) Find the additive inverse of each of its elements.	**JULY 2022**	**07**
INTEGRAL DOMAIN (ALGEBRAIC STRUCTURES WITH TWO BINARY OPERATIONS)			
1	Show that $(R, +, \times)$ is an integral domain, where $R = \{a + b\sqrt{11} \mid a, b \in I\}$.	**MAY 2025**	**07**
2	Show that $(R, +, \times)$ is an integral domain, where $R = \{a + b\sqrt{5} \mid a, b \in \mathbb{Z}\}$.	**JULY 2024** **JULY 2023**	**07** **07**
INTEGRAL DOMAIN & FIELD (ALGEBRAIC STRUCTURES WITH TWO BINARY OPERATIONS)			
1	Show that $R = \{a + b\sqrt{2};\ a, b \in I\}$ is an integral domain but not a field under the binary operations $+$ (usual addition) and $\cdot$ (usual multiplication).	**NOV 2025**	**07**
FIELD (ALGEBRAIC STRUCTURES WITH TWO BINARY OPERATIONS)			
1	Consider the set $(C, +, \cdot)$, where C is the set of complex numbers and $+$ and $\cdot$ are ordinary addition and multiplication operation. Show that $(C, +, \cdot)$ is a field.	**NOV 2025**	**07**

2	Show that the set of all matrices of the form $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$ where $a, b \in R$ is a field with respect to matrix addition and matrix multiplication.	**DEC 2022**	**07**
PRINCIPAL IDEAL			
1	Define Principal Ideal. Find all the principal Ideal of the ring $[\{0, 1, 2, 3, 4, 5\}, +_6, \times_6]$.	**DEC 2022**	**07**

MODULE 4 – RELATIONS

SR. NO.	TOPIC / QUESTION	YEAR	MARKS
RELATION AS SETS			
1	Let L denotes the relation "less than or equal to" and D denotes the relation "divides". Where xDy means "x divides y". Both L and D are defined on the set $\{1, 2, 3, 6\}$. Write L and D as sets, find $L \cap D$.	**DEC 2021**	**04**
TYPES OF A RELATION			
1	Explain types of a Relation with a suitable example.	**SEP 2021**	**03**
PROPERTIES OF BINARY RELATIONS: REFLEXIVE, SYMMETRIC, TRANSITIVE, ANTI-SYMMETRIC, IRREFLEXIVE			
1	Check whether the relation R defined in the $\{1, 2, 3, 4, 5, 6\}$ as $R = \{(a, b) : b = a + 2\}$ is reflexive, symmetric or transitive.	**MAY 2025**	**04**
2	For the relation $R = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$ on the set $A = \{1, 2, 3, 4\}$. Check whether it is reflexive, symmetric, anti-symmetric, transitive.	**NOV 2024**	**04**
3	Determine the relation $\leq$ (less than or equal) on the set $\mathbb{Z}$ of integers are reflexive, symmetric, anti-symmetric, transitive.	**JULY 2024**	**04**
4	For the relation $\{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$ on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is anti-symmetric, and whether it is transitive. (Justify your answer if the property is not satisfied)	**JULY 2022**	**03**
5	Check whether the relation R defined by $R = \{(a, b) : a \leq b^3, a \text{ and } b \text{ are real numbers}\}$ is reflexive, symmetric or transitive.	**MAY 2025**	**04**
6	Check whether the relation R defined in the $\{1, 2, 3, 4, 5, 6\}$ as $R = \{(a, b) : b = a + 1\}$ is reflexive, symmetric or transitive.	**JULY 2023**	**04**
7	Given $S = \{1, 2, 3, \ldots, 10\}$ and a relation R on S, where $R = \{\langle x, y\rangle \mid x + y = 10\}$. What are the properties of relation R?	**DEC 2021**	**03**
8	Let N be the set of natural numbers. Let R be a relation in N defined by xRy if and only if $x + 3y = 12$. Examine the relation for (i) reflexive (ii) symmetric (iii) transitive.	**FEB 2021**	**03**
9	Determine the relation $\parallel$ (parallel) on the set L of lines in the plane are reflexive, symmetric, anti-symmetric, transitive, irreflexive.	**FEB 2024**	**04**

PARTITION OF A SET			
1	Which of these collections of subsets are partitions of $\{1,2,3,4,5,6\}$? Justify your answer if it is not a partition. (i) $\{1,2\}$, $\{2,3,4\}$,$\{4,5,6\}$ (ii) $\{1,4,5\}$, $\{2,6\}$	**JULY 2022**	**04**
2	Define Partition of a Set. Let $A = \{1,2,3,4,5\}$ and $R = \{(1,2),(1,1),(2,1),(2,2),(3,3),(4,4),(5,5)\}$ be an equivalence relation on A. Determine the partition for R^{-1}, if it is an equivalence relation.	**DEC 2022**	**03**
3	Let R be a relation defined in $A = \{1,2,3,5,7,9\}$ as $R = \{(1,1),(1,3),(1,5),(1,7),(2,2),$ $(3,1),(3,3),(3,5),(3,7),(5,1),(5,3),(5,5),(5,7),(7,1),$ $(7,3),(7,5),(7,7),(9,9)\}$. Find the partitions of A based on the equivalence relation R.	**FEB 2021**	**03**
EQUIVALENCE RELATION			
1	Define equivalence relation. Let $X = \{1,2,3,4,5\}$, $R = \{\langle x,y\rangle \ / \ x$ is divisible by $y\}$. Check whether the relation is an equivalence relation.	**FEB 2024**	**03**
2	Show that the relation $x \equiv y \pmod m$ defined on the set of integers I is an equivalence relation.	**OCT 2020**	**03**
3	If R is the relation defined in Q (the set of rational numbers) by $R = \left\{\left(\frac{a}{b},\frac{c}{d}\right) \in Q \times Q : ad = bc\right\}$ then show that R is an equivalence relation.	**MAY 2025**	**04**
4	Let $X = \{1,2,3,\ldots,7\}$ and $R = \{\langle x,y\rangle \mid x - y$ is divisible by $3\}$. Show that R is an equivalence relation on X. Draw the graph of R.	**SEP 2021** **DEC 2021**	**04** **07**
5	Determine whether the relation with the directed graph shown is an equivalence relation. 	**JULY 2022**	**04**
EQUIVALENCE CLASSES			
1	Define equivalence class. Let R be the relation on the set of integers I defined by $x - y$ is an even integer, find the disjoint equivalence classes.	**OCT 2020**	**03**

2	Define equivalence class generated by an element $x \in X$. Let Z be the set of integers and let R be the relation called "congruence modulo 3" defined by $R = \{\langle x, y\rangle \mid x \in Z \wedge y \in Z \wedge x - y$ is divisible by $3\}$ Determine the equivalences classes generated by the element of Z.	**DEC 2021**	**07**
MAXIMAL COMPATIBILITY BLOCK			
1	Find out maximal compatibility blocks of following simplified graph of digraph and write its relation matrix.  	**FEB 2024**	**04**
COMPOSITE RELATION			
1	For the following relations on the set of real numbers, $R_1 = \{(a, b) \in R^2 \mid a \geq b\}$, $R_2 = \{(a, b) \in R^2 \mid a \leq b\}$, $R_3 = \{(a, b) \in R^2 \mid a \neq b\}$, find (1) $R_1 \oplus R_3$ (2) R_2 o R_3.	**JULY 2022**	**04**
2	If $R = \{(a, b) : \lvert a - b\rvert = 1\}$ and $S = \{(a, b) : a - b$ is even$\}$ be two relations on $A = \{1, 2, 3, 4\}$. Then (I) Find matrices of R and S (II) Find digraph of R and S (III) Find the relation RS.	**JULY 2023**	**07**
INVERSE AND COMPLEMENT OF RELATION			
1	Consider the relation R defined on $A = \{1, 2, 3\}$ whose matrix representation is given below. Determine its inverse R^{-1} and complement R'. $M_R = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	**DEC 2022**	**04**
GRAPH OF RELATION			
1	Let $S = \{1, 2, 3, 4\}$ and $R = \{(1,1), (1,4), (2,2), (2,3), (3,2), (3,3), (4,1), (4,4)\}$. Draw the graph of R and hence write partition of S.	**JULY 2024**	**03**
2	Let $X = \{1, 2, 3, 4\}$ and $R = \{\langle x, y\rangle / x > y\}$. Draw the graph of R and also give its matrix.	**FEB 2024**	**03**

RELATION MATRIX			
1	If $A = B = \{1, 2, 3, 5, 6, 10, 15, 30\}$ and relation R defined as a/b, where $a \in A$ and $b \in B$. Find the relation matrix.	**NOV 2025**	**03**
2	Let $A = \{1, 2, 3, 4\}$ and the relation $R = \{\{(1, 1), (1, 4), (4, 1), (4, 4), (2, 2), (2, 3), (3, 2), (3,3)\}$ on A. Write the matrix of R and check whether the relation is an equivalence.	**NOV 2024**	**03**
MATRIX OF COMPOSITE RELATION			
1	Given the relation matrices M_R and M_S, find $M_{R\circ S}$, $M_{\tilde{R}}$, $M_{\tilde{S}}$, $M_{\widetilde{R\circ S}}$, and show that $M_{\widetilde{R\circ S}} = M_{\tilde{S}} \circ M_{\tilde{R}}$. $\mathrm{M_R} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and $\mathrm{M_S} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$.	**JULY 2024**	**07**
MATRICES OF UNION AND INTERSECTION OF RELATIONS			
1	Suppose that the relations R_1 and R_2 on a set A are represented by the matrices $M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ and $M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$. What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$?	**JULY 2022**	**03**
TRANSITIVE CLOSURE BY WARSHALL'S ALGORITHM			
1	Find transitive closure by Wars hall's Algorithm if $A = \{1, 2, 3, 4, 5\}$ and $R = \{(1,4), (2,1), (2,5), (2,4), (4,3), (5,3), (3,2)\}$	**MAY 2025**	**07**
2	Let the relation $R = \{(1, 2), (2, 3), (3, 3)\}$ on $A = \{1, 2, 3\}$. Find the transitive closure of R.	**NOV 2024**	**03**
3	Find the transitive closure of $R = \{(1,2), (3,4), (4,5), (4,1), (1,1)\}$ where $A = \{1, 2, 3, 4, 5\}$.	**SEP 2021**	**04**
4	Use Warshall's Algorithm to find the transitive closure of the relation $R = \{(1,4) , (2,1) , (2,3) , (3,1) , (3,4), (4,3)\}$ on $\{1, 2, 3, 4\}$.	**JULY 2022**	**04**
5	Let $A = \{1, 2, 3, 4\}$. For the relation R whose matrix is given, find the matrix of the transitive closure by using Warshall's algorithm. $M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	**DEC 2021**	**07**
6	Find transitive closure by Warshall's Algorithm if $A = \{1, 2, 3, 4, 5\}$ and $R = \{(1,2) , (3,4) , (4,5) , (4,1) , (1,1)\}$.	**JULY 2023**	**04**

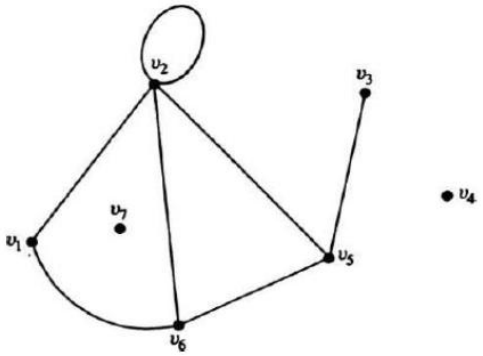
MODULE 4 – PARTIAL ORDERING

SR. NO.	TOPIC / QUESTION	YEAR	MARKS
PARTIAL ORDER RELATION			
1	Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 18, 24\}$ be ordered by the relation x divides y. Show that the relation is partial ordering.	**NOV 2025**	**03**
2	Explain Partially order relation with example.	**DEC 2022** **NOV 2025**	**03** **01**
PARTIALLY ORDERED SET (POSET)			
1	Explain Partially ordered set (Poset) with example.	**SEP 2021** **NOV 2025**	**01** **01**
HASSE DIAGRAM			
1	Let $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 18, 24\}$ be ordered by the relation x divides y. Draw the Hasse diagram.	**NOV 2025**	**04**
2	Explain Hasse diagram with example.	**NOV 2025**	**01**
3	Draw Hasse diagram for $P = \{1, 2, 3, 5, 11, 13, 17, 19, 23\}$ and $\leq$ is a relation such that $x \leq y \Leftrightarrow x \mid y$.	**MAY 2025**	**03**
4	Draw Hasse diagram of $\langle S_{30}, D \rangle$. Prove that $\langle S_{30}, D \rangle$ is a lattice, where D is the relation of "division" in $\mathbb{N}$ such that for any $a, b \in \mathbb{N}$, aDb if and only if a divides b and S_n, $(n \in \mathbb{N})$ is the set of all divisors of n.	**NOV 2024**	**07**
5	Define Lattice. Draw the Hasse diagram of (S_{12}, D), where D is the relation of "division" in $\mathbb{N}$ such that for any $a, b \in \mathbb{N}$, aDb iff a divides b and S_{12} is the set of all divisors of 12.	**JULY 2024**	**04**
6	Draw the Hasse diagram representing the partial ordering $\{(a, b) \mid a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$.	**FEB 2021**	**04**
7	Draw Hasse Diagram for the poset (S_{30}, D) where S_{30} is the set of divisors of 30 and D is the relation divides.	**DEC 2022**	**04**
8	Let A be the set of factors of particular positive integer m and $\leq$ be the relation divides, that is $\leq = \{\langle x, y \rangle \mid x \in A \wedge y \in A \wedge (x \text{ divides } y)\}$ Draw the Hasse diagrams for $a)\ m = 45\ \ b)\ m = 210$.	**DEC 2021**	**07**
9	Draw the Hasse diagram of the set $\{1, 3, 9, 18\}$ under partial order relation 'divides' and indicate those which are chains.	**SEP 2021**	**03**

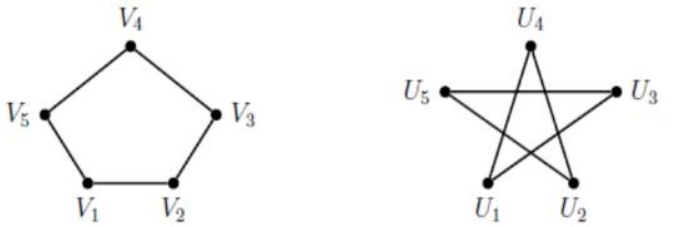
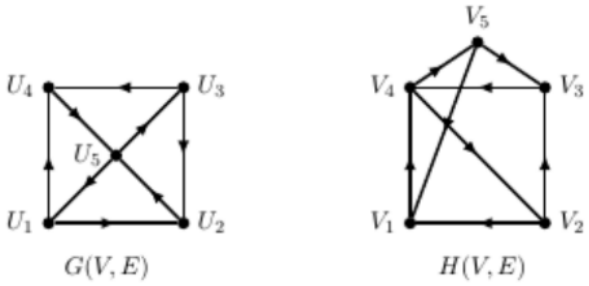
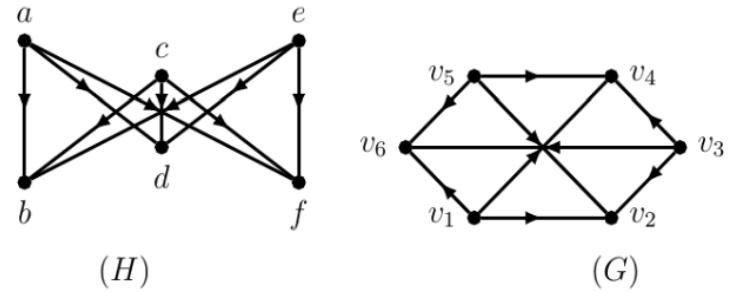
10	Let A be a given finite set and $\rho(A)$ its power set. Let $\subseteq$ be the inclusion relation on the elements of $\rho(A)$. Draw the Hasse diagram of $\langle \rho(A), \subseteq \rangle$ for $A = \{a, b, c\}$.	**DEC 2021**	**03**
11	Draw the Hasse diagram for the partial ordering $\{(A, B) \mid A \subseteq B\}$ on the power set $P(S)$ where $S = \{a, b, c\}$.	**OCT 2020**	**04**
12	Let A be a given finite set and $\rho(A)$ its power set. Let $\subseteq$ be the inclusion relation on the elements of $\rho(A)$. Draw the Hasse diagram of $\langle \rho(A), \subseteq \rangle$ for $A = \{a, b, c, d\}$.	**DEC 2021** **FEB 2024**	**04** **04**
MAXIMAL AND MINIMAL ELEMENTS LEAST & GREATEST ELEMENTS			
1	Define Lattice. Consider a poset $(\{3, 5, 9, 15, 24, 45\}, \mid)$ where $\mid$ denotes "divides" is a lattice. Then **(i)** draw its Hasse diagram **(ii)** Find its maxima, minima, greatest and least elements when they exist **(iii)** Find maxima, minima, greatest and least elements of the set $M = \{3, 9, 15\}$ when they exist.	**MAY 2025**	**07**
2	Draw the Hasse diagram for the poset $(\{2, 4, 5, 10, 12, 20, 25\}, \mid)$. Hence, find the maximal and minimal elements.	**JULY 2022**	**03**
3	Define Lattice and draw the Hasse diagram representing the partial ordering $\{(A, B) : A \subseteq B\}$ on the power set $P(S)$ where $S = \{1, 2, 3\}$. Find the maximal, minimal, greatest and least elements of the poset.	**JULY 2023**	**07**
GREATEST LOWER BOUND (GLB) (INFIMUM) & LEAST UPPER BOUND (LUB) (SUPREMUM)			
1	Let $X = \{2, 3, 6, 12, 24, 36\}$ and the relation $\leq$ be such that $x \leq y$ if x divides y. Find **(i)** lower bound of {3,6} **(ii)** upper bound of {3,6} **(iii)** GLB of {3,6} if exists **(iv)** LUB of {3,6} if exists.	**NOV 2024**	**04**
2	Let the POSET $(\rho(A), \leq)$ where $A = \{a, b, c\}$, relation is subset. Find (i) upper bound of $\{\{\}, \{a\}, \{c\}\}$ (ii) GLB of $\{\{\}, \{a\}, \{c\}\}$, if exists (iii) LUB of $\{\{\}, \{a\}, \{c\}\}$, if exists	**JULY 2024**	**03**
3	Draw the Hasse diagram for the poset $(\{2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72\}, \mid)$. Hence, find $glb(\{60, 72\})$ and all maximal elements.	**JULY 2022**	**03**
LATTICE			
1	Let n be a positive integer, S_n be the set of all divisors of n. Let D denote the relation of division. Draw the lattices for (i) $n = 24$ (ii) $n = 30$.	**NOV 2025**	**03**

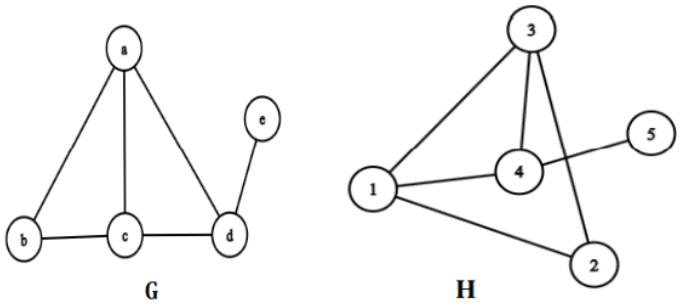
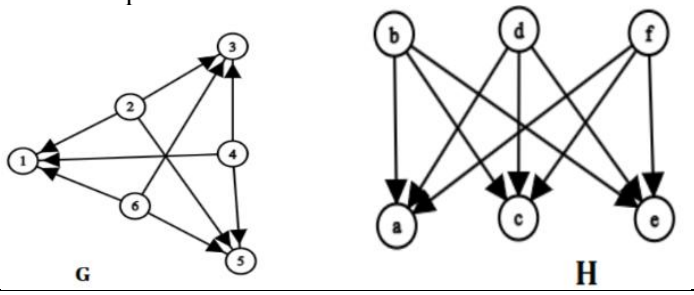
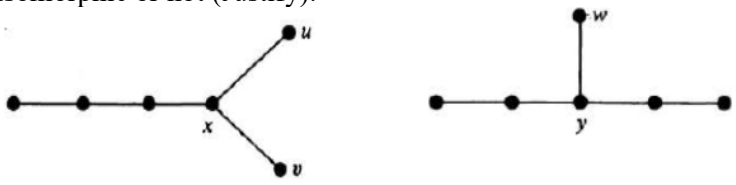
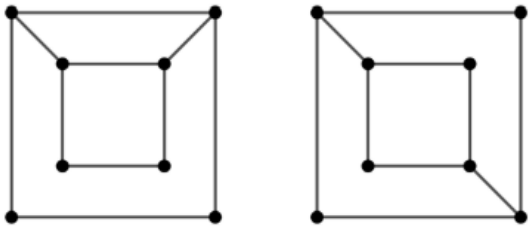
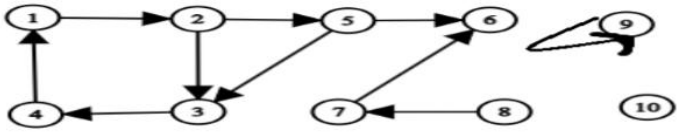
2	Define Lattice.	**MAY 2025**	**01**
3	Let A be the set of factors of positive integer m and relation is divisibility on A. For $m = 45$, show that POSET $(A, \leq)$ is a Lattice.	**SEP 2021**	**07**
4	Define lattice. Determine whether POSET $\{\{1, 2, 3, 4, 5\}\ ;\ |\}$ is a lattice.	**FEB 2021** **OCT 2020**	**04** **04**
LATTICE INEQUALITIES			
1	Let $\langle L, \leq \rangle$ be a lattice. Show that for a, b, $c \in L$, following inequalities holds. $a \oplus (b * c) = (a \oplus b) * (a \oplus c)$ $a * (b \oplus c) = (a * b) \oplus (a * c)$	**DEC 2021**	**07**
2	Show that in a lattice if $a \leq b \leq c$ then **(i)** $a \oplus b = b * c$ and **(ii)** $(a * b) \oplus (b * c) = b = (a \oplus b) * (a \oplus c)$	**MAY 2025** **JULY 2023**	**04** **04**
3	Let $\langle L, \leq \rangle$ be a lattice. Show that for $a, b, c \in L$, following inequalities holds. (i) $a \oplus (b * c) \leq (a \oplus b) * (a \oplus c)$ and (ii) $a * (b \oplus c) \geq (a * b) \oplus (a * c)$	**JULY 2024**	**07**
BOUNDED, DISTRIBUTIVE AND COMPLEMENTED LATTICES			
1	Define: Bounded, Distributive and Complemented Lattices.	**SEP 2021**	**03**
BOOLEAN ALGEBRA			
1	Define Boolean algebra.	**SEP 2021**	**01**
2	Prove that $\langle S_{30}, D \rangle$ is a Boolean algebra.	**FEB 2024**	**07**

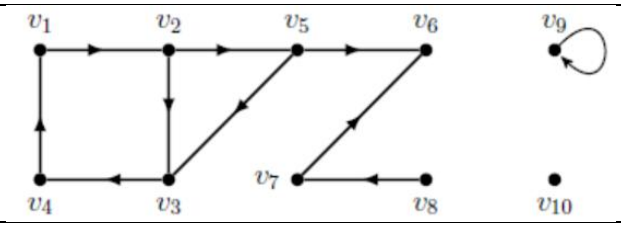
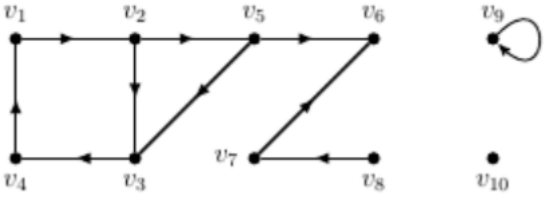
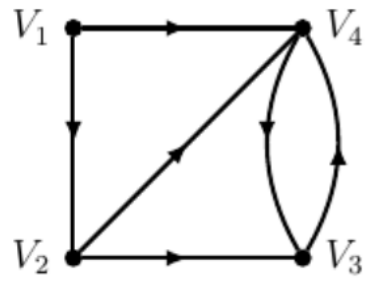
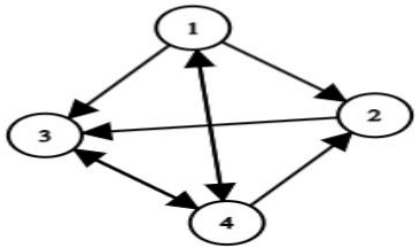
MODULE 5 – GRAPHS & TREES

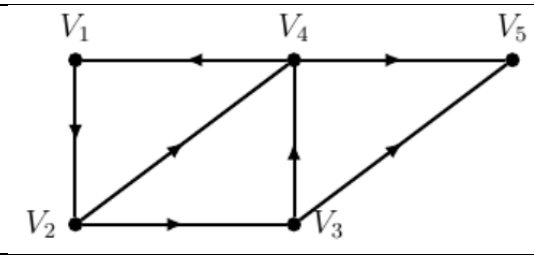
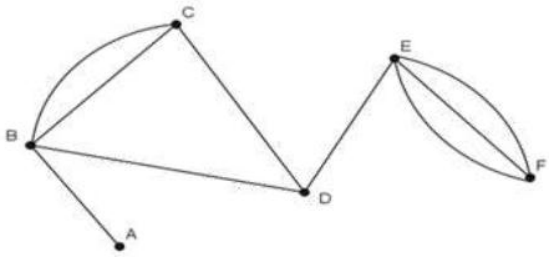
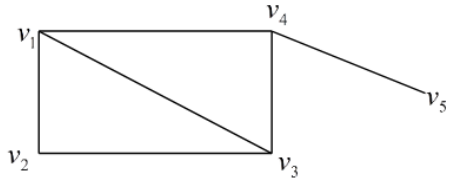
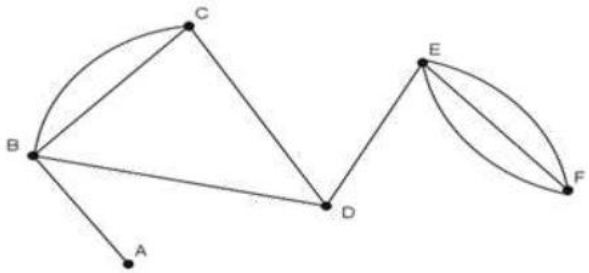
SR. NO.	TOPIC / QUESTION	YEAR	MARKS
BASIC DEFINITIONS			
1	Define simple graph, degree of a vertex, finite and infinite graphs and complete graph.	**NOV 2025**	**04**
2	Define Isolated vertex, Pendent vertex and Size of a graph.	**JULY 2023**	**03**
3	Define self-loop, adjacent vertices and a pendant vertex.	**FEB 2021**	**03**
4	Draw the graph with 4 nodes and 7 edges.	**NOV 2025**	**03**
5	Define the terms: Simple Graph, Multi-Graph, Weighted Graph, Degree of a vertex, in degree and out degree of a vertex. Illustrate each with an example.	**DEC 2022**	**07**
6	Define indegree and outdegree of a vertex in a graph with example.	**FEB 2021**	**03**
7	Define elementary cycle, loop, tree and pendent vertex.	**NOV 2024**	**04**
8	Define connected graph, Boolean matrix and Euler path.	**MAY 2025**	**03**
9	Define simple graph, degree of a vertex and complete graph.	**OCT 2020**	**03**
10	Define Isolated node, Binary tree and Regular graph.	**JULY 2024**	**03**
11	Define Isolated vertex and Null graph. Identify Isolated vertex/vertices from the following graph. 	**JULY 2022**	**03**
12	Define path and circuit of a graph by drawing the graphs.	**FEB 2021** **JULY 2022**	**04** **03**
13	Define cycle and walk.	**JULY 2023**	**02**
14	Define null graph, centre of a graph and complete graph.	**FEB 2024**	**03**
15	Define centre of a graph.	**FEB 2021**	**02**

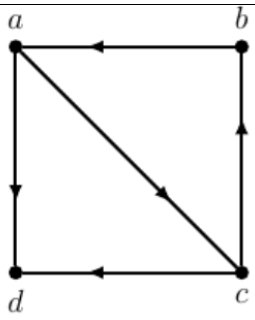
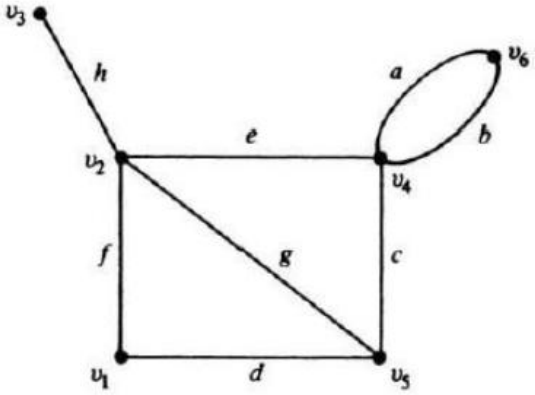
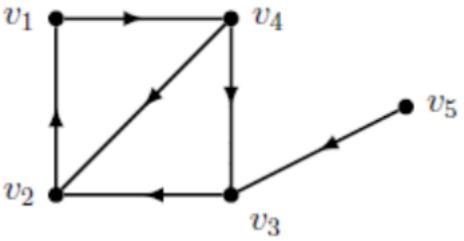
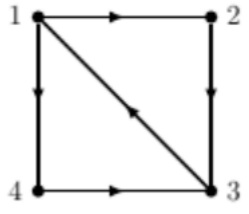
16	Explain the following terms with proper illustrations. a) Directed graphs b) Simple and elementary path c) Reachability of a vertex d) Connected graph	**DEC 2021**	**07**
17	Define Cyclic graph, Null graph and strongly connected graph.	**SEP 2021**	**03**
DEGREE, IN-DEGREE, OUT-DEGREE, TOTAL DEGREE OF A NODE			
1	Show that sum of in-degrees of all the nodes of simple digraph is equal to the sum of out-degrees of all the nodes and this sum equal to the number of edges in it.	**DEC 2021**	**07**
SUBGRAPHS			
1	Define vertex disjoint and edge disjoint subgraphs by drawing the relevant graphs.	**OCT 2020**	**04**
2	Define complement of a subgraph by drawing the graphs.	**FEB 2021**	**04**
HANDSHAKING THEOREM (DEGREE SUM FORMULA)			
1	Does there exist a graph with 20 edges and each vertex of degree 3?	**MAY 2025**	**03**
2	A graph G has 15 edges, 3 vertices of degree 4 and other vertices of degree 3. Find the number of vertices in G.	**JULY 2024** **OCT 2020**	**03** **03**
3	Let G be a graph with n vertices and m edges such that vertices have degree k or $k+1$. Prove that if G has N_k vertices of degree k and N_{k+1} vertices of degree $k+1$ then $N_k = (k+1)n - 2m$.	**JULY 2023**	**04**
4	Find the number of edges in G if it has 5 vertices each of degree 2.	**FEB 2021**	**03**
5	How many nodes are necessary to construct a graph with exactly 8 edges in which each node is of degree 2?	**SEP 2021**	**03**
6	Prove that any graph has even number of odd vertices.	**JULY 2023**	**04**
7	Prove that there are always an even number of vertices of odd degree in a graph.	**OCT 2020**	**03**
SPECIAL TYPES OF GRAPHS (REGULAR GRAPHS & COMPLETE GRAPHS & COMPLETE BIPARTITE GRAPHS)			
1	Explain the regular graph and complete bipartite graph with example(s).	**NOV 2025**	**04**

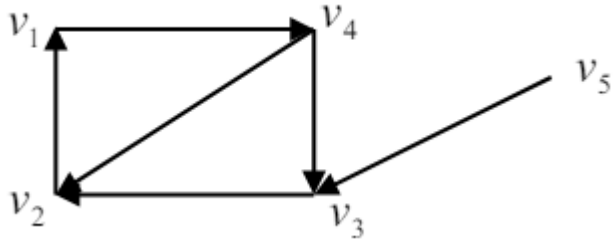
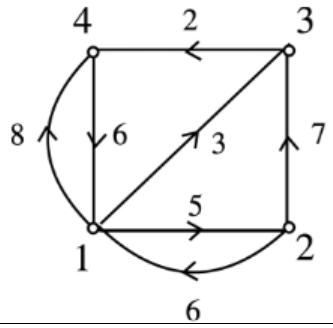
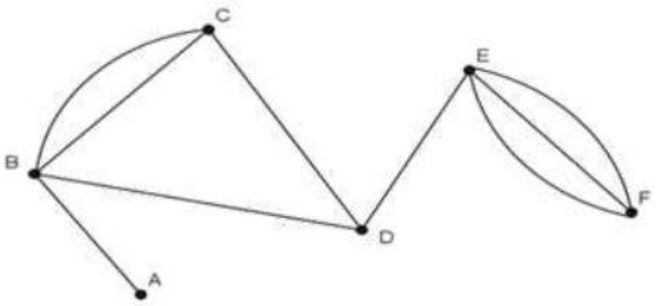
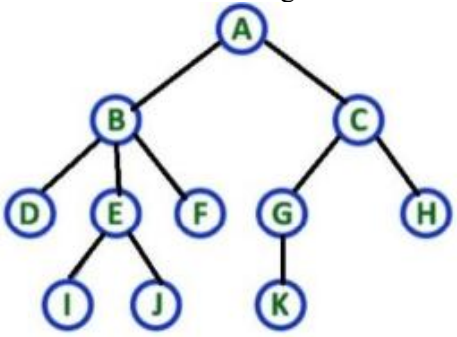
2	Draw the following graphs: (i) $K_{3,4}$ (ii) $K_{4,4}$ (iii) $K_{5,4}$ (iv) K_5	**NOV 2025**	**04**
3	Define complete graph and find order and size of the graph K_{2025}.	**MAY 2025**	**03**
4	**(i)** Draw a complete bipartite graph which is not regular. **(ii)** Draw a graph which is regular but not bipartite.	**MAY 2025**	**04**
5	Find the number of edges in a r-regular graph with n vertices.	**NOV 2024**	**03**
6	Draw a graph which is regular but not bipartite.	**SEP 2021**	**04**
7	Find the number of edges in a r - regular graph with n - vertices.	**OCT 2020**	**03**
8	Does a 3 - regular graph with 5 vertices exist?	**FEB 2021**	**03**
9	Does there exist a 4 - regular graph with 6 vertices? If so, construct the graphs.	**JULY 2023**	**04**
ISOMORPHIC GRAPHS			
1	Check whether the following graphs are isomorphic or not. 	**NOV 2024**	**04**
2	Check whether the following graphs are isomorphic or not. 	**JULY 2024**	**04**
3	Check whether the following graphs are isomorphic. 	**FEB 2024**	**04**

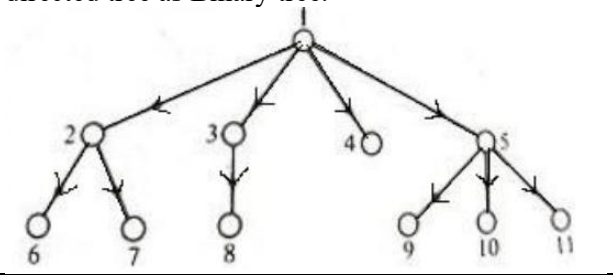
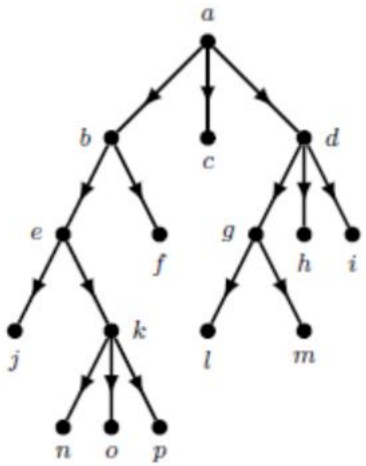
4	Define Isomorphic Graphs. Determine whether the following graphs are isomorphic or not.  G H	**JULY 2023**	**07**
5	Define Isomorphic graphs. Check whether the following graphs are isomorphic.  G H	**MAY 2025**	**07**
6	Define Isomorphic Graphs and verify the following graphs are Isomorphic or not (Justify).  Graph -1 Graph-2	**JULY 2022**	**04**
7	Define Isomorphic Graphs. Verify the following graphs are Isomorphic or not (Justify). 	**SEP 2021**	**07**
REACHABLE SET OF A GIVEN NODE & NODE BASE			
1	Find a Node base for the following graph:  	**MAY 2025**	**04**
2	Find reachable set of each node of the given digraph.	**NOV 2024**	**04**

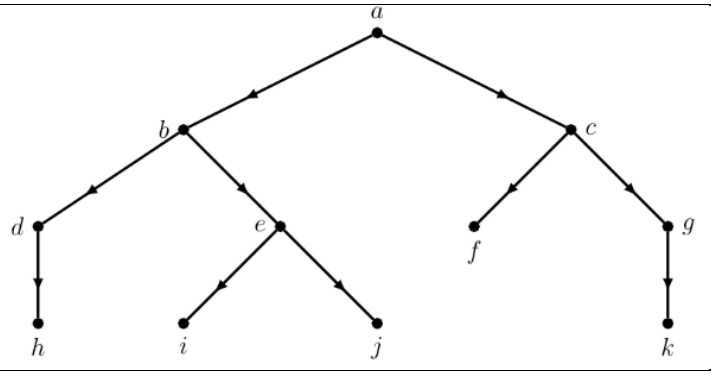
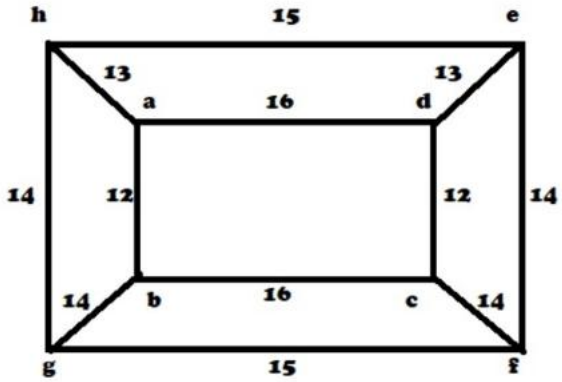
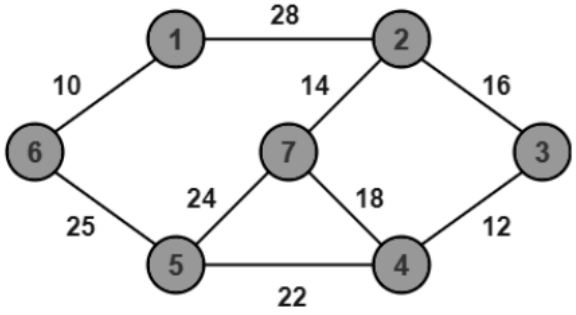
			
3	Find all the node base of the given digraph. Also find $d(V_3, V_6)$, $d(V_6, V_3)$. 	**JULY 2024**	**04**
4	Find reachable set of each node of the given digraph. Also find $d(V_1, V_3)$, $d(V_3, V_1)$. 	**FEB 2024**	**04**
5	Find reachable set of all the vertices and node base of following graph. 	**JULY 2023**	**03**
WEAKLY CONNECTED, STRONGLY CONNECTED & UNILATERALLY CONNECTED (STRONG, WEAK & UNILATERAL COMPONENTS OF A GRAPH)			
1	Define the following by drawing graphs (i) weak component (ii) unilateral component (iii) strong component.	**OCT 2020**	**04**
2	Define Strong, unilateral, weak component. Also Find strong, unilateral, weak component from the given digraph.	**FEB 2024**	**07**

			
CUT VERTEX			
1	Define cut vertex. List out all the cut vertices of the graph given in figure. 	**DEC 2022**	**03**
MATRIX REPRESENTATION OF GRAPH ADJACENCY MATRIX			
1	Define adjacency matrix and find the same for 	**FEB 2021**	**04**
2	Define adjacency matrix and path matrix of a graph. Find out adjacency matrix for the graph given in figure. 	**DEC 2022**	**04**
PATH (REACHABILITY) MATRIX OF A GRAPH			
1	Find the path matrix using adjacency matrix of the following graph.	**FEB 2024**	**04**

			
INCIDENCE MATRIX			
1	Define incidence Matrix of a Graph. Find incidence matrix for the following graph. 	**JULY 2022**	**04**
WARSHALL'S ALGORITHM TO PRODUCE PATH MATRIX			
1	Define adjacency matrix. Use Warshall's algorithm to obtain path matrix from the adjacency matrix of 	**NOV 2024**	**07**
2	Define path matrix. Use Warshall's algorithm to obtain path matrix from the adjacency matrix of following graph 	**JULY 2024**	**07**
3	Use Warshall's algorithm to obtain path matrix from the adjacency matrix of	**OCT 2020**	**04**

			
SHORTEST PATH			
1	Find the shortest path between each pair of vertices for a simple digraph using Warshall's Algorithm. 	**SEP 2021**	**04**
EULERIAN GRAPH, PLANAR GRAPH & EULER'S FORMULA			
1	Define Eulerian graph, Planar Graph. Justify whether the graph given in figure is Planar or not using Euler's formula. 	**DEC 2022**	**04**
BASIC DEFINITIONS OF GRAPHS AND TREES			
1	Define Subtree and Degree of a Node. Determine degree of each node for the following tree. 	**JULY 2022**	**07**

2	Define tree and complete graph.	**SEP 2021**	**02**
3	Define radius of a tree. Define tree.	**FEB 2021** **JULY 2023**	**02** **01**
BASIC DEFINITIONS & REPRESENTATIONS OF A TREE & BINARY TREE			
1	Define tree and root. Also draw a tree with 10 vertices which has vertices either of degree 1 or of degree 3. Is it possible to draw a same type of tree with 11 vertices?	**NOV 2025**	**07**
2	Find total number of vertices of a full ternary tree with n levels.	**MAY 2025**	**03**
REPRESENTATION OF A TREE AS BINARY TREE			
1	Define: M-ary Tree and Binary Tree. Represent the following directed tree as Binary tree. 	**JULY 2022**	**07**
TREE TRAVERSAL: PRE-ORDER, IN-ORDER, POST-ORDER TRAVERSAL			
1	In which order does a pre-order, in-order and post-order traversal visit the vertices of the ordered rooted tree shown in figure? 	**NOV 2024**	**07**
2	Define tree. In which order does a pre-order, in-order and post-order traversal visit the vertices of the ordered rooted tree shown in figure?	**FEB 2024**	**07**

			
MINIMUM SPANNING TREE (PRIM'S ALGORITHM / KRUSKAL'S ALGORITHM)			
1	Find minimum spanning tree for the weighted graph given below. 	**NOV 2025**	**07**
2	Define Binary tree, Spanning tree, Minimal Spanning tree. Find the minimal spanning tree for the weighted graph given in figure. 	**DEC 2022**	**07**
DEGREE SUM FORMULA IN TREES (HANDSHAKING LEMMA FOR TREES)			
1	A tree T has 4 vertices of degree 2, 3 vertices of degree 3, 1 vertex of degree 4. Find the number of pendant vertices in the tree T.	**NOV 2024**	**03**
2	A tree T has 3 vertices of degree 4, 3 vertices of degree 3. Find the number of pendant vertices in tree T.	**FEB 2024**	**03**
3	A tree T has 4 vertices of degree 2, 4 vertices of degree 3, 2 vertices of degree 4. Find the number of pendant vertices in T.	**OCT 2020**	**04**

PROPERTIES OF TREES (EDGES–VERTICES RELATIONSHIP)			
1	Prove that a tree with n vertices has $n-1$ edges.	**JULY 2024**	**04**
2	Define tree and root. Also prove that tree with n vertices has $n-1$ edges.	**DEC 2021**	**07**
3	Define tree. Prove that if a graph G has one and only one path between every pair of vertices then G is a tree.	**FEB 2021** **OCT 2020**	**04** **04**
BINARY TREE RECONSTRUCTION FROM TRAVERSALS (PRE-ORDER, IN-ORDER, POST-ORDER)			
1	Draw binary trees whose post-order produced the string d-e-c-g-j-h-f-b-l-n-q-r-p-m-k-a and pre-order produced the string a-b-d-h-e-i-j-c-f-g-k and in-order produced the string h-d-b-i-e-j-a-f-c-k-g.	**JULY 2024**	**07**
2	Draw a binary tree whose post-order produced the string $d-g-e-b-i-j-h-f-c-a$ and pre-order produces the string $j-b-a-c-d-i-h-e-g-f$.	**JULY 2023**	**04**
3	Explain Post order traversal. Given the post order and in order traversal of a binary tree, draw the unique binary tree. Post order: $d\ e\ c\ f\ b\ h\ i\ g\ a$ In order: $d\ c\ e\ b\ f\ a\ h\ g\ i$	**DEC 2022**	**07**
BINARY SEARCH TREE (BST)			
1	Form a binary search tree for the data 16 , 24 , 7 , 5 , 8 , 20 , 40 , 3.	**DEC 2022**	**04**
HUFFMAN CODING (OPTIMAL BINARY PREFIX CODES)			
1	For the following set of weights construct an optimal binary prefix code. For each weight in the set, give corresponding code word. 5 , 7, 8 , 15 , 35 , 40	**SEP 2021**	**07**